PINGLIFTS FITNESS

The Only Muscle Building Cheat Sheet You Ever Need

Training. Nutrition. Supplements. No guessing.



Introduction — Stop Guessing. Start Progressing.

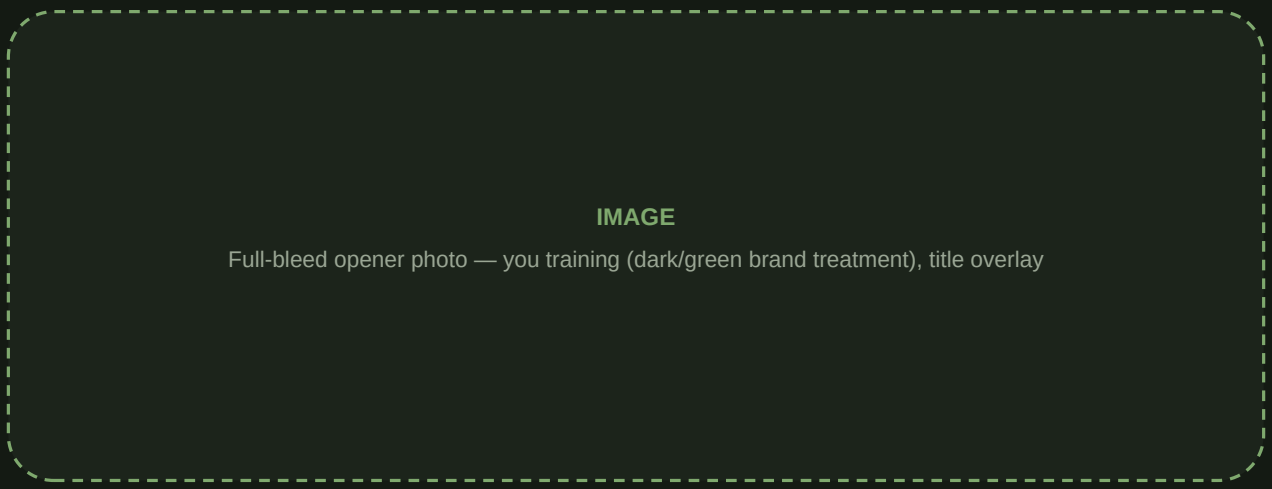
Let's be honest about how most people train.

They bounce between whatever program their favorite influencer is running this month. They chase the pump, judge workouts by soreness, and swap exercises before anything has time to work. They eat "clean" without knowing what that's supposed to do, buy supplements that promise what training and food already provide, and then wonder why the mirror hasn't changed in two years.

Building muscle isn't complicated. It's just misunderstood — buried under bro-science, guru wars, and products that need you confused to sell.

This guide is my fix for that. Three parts:

- **Part 1 — Training.** What actually builds muscle, where the experts genuinely disagree (and how to think about it), and the programming system that ties it together: progression, volume, exercise selection, splits, and recovery.
- **Part 2 — Nutrition.** Calories, protein, carbs and fat, and meal timing — including the parts most guides skip, like what to do when the scale stalls and how to eat around a real life.
- **Part 3 — Supplementation.** The short, honest list of what works, how to read a label so you stop buying junk, and everything you can safely ignore.



How to read this

Straight through if you're newer — each part builds on the last. Skim the decision tree in Chapter 6 if you only have ten minutes; it's the distilled version of the whole training half.

One promise on tone: I won't pretend every question has a settled answer. Where smart coaches disagree, I'll show you the disagreement and tell you what I'd do about it. You deserve the real map, not a tidy story.

Let's get to work.

Part 1 — Training

Chapter 1 — What Actually Builds Muscle

Forget the pump. Forget the burn. Forget how sore you were for three days after leg day.

Those things feel productive, and that's exactly the problem — feeling productive and being productive are not the same thing in the gym.

One driver: mechanical tension

Mechanical tension — the pulling force your muscle fibers experience as they contract against resistance — is the primary, and arguably the only, driver of muscle growth. This is about as close to undisputed as exercise science gets.

The older textbook model named three drivers: mechanical tension, metabolic stress (the pump/burn), and muscle damage (soreness). Research has since gutted the other two:

- **Muscle damage doesn't grow muscle.** Studies that tracked damage against long-term growth found no relationship — and damaged muscle actually produces *less* force, because inflammation makes your nervous system throttle recruitment.
- **The pump doesn't grow muscle.** Shorter rest periods create more metabolic stress but produce *less* growth; the burn is a side effect, not a signal.

Damage and stress are passengers. Tension is driving.

IMAGE

Three-panel myth graphic — "Pump NO", "Soreness NO", "Tension YES" with one-line explanations

Tension is a product of effort

Creating enough tension comes down to training close enough to failure that the largest, highest-growth-potential motor units have to fire. The practical unit here is **RIR — Reps In Reserve**. A set ending at 1 RIR means: if you had gone all-out, you could have completed exactly one more good rep.

The evidence-based sweet spot for most sets is around **1–3 RIR** — hard enough to generate full recruitment, without drowning yourself in the fatigue that failure on every set creates. The last five-ish reps before failure are where the growth-stimulating work happens.

How do you know you're actually near failure? The reliable sign isn't pain — it's **rep speed and range of motion involuntarily dropping** even while you push maximally. Bar slowing, ROM shortening, grinding: that's proximity.

No rep range is magic

Here's a myth worth killing early: the "6–12 hypertrophy zone." In reality, **5 to 30 reps build essentially equal muscle — provided the set gets within a couple of reps of failure**. Twenty-rep sets and six-rep sets both grow you if both are genuinely hard.

So why do most programs live in 6–12 (or 6–15)? Practicality: very heavy sets beat up joints and make accumulating volume harder; very long sets become cardio-limited and fatiguing. It's a logistics choice, not biology. Heavy weight does not "build mass" while light weight "tones" — tone is a superstition.

IMAGE

Rep-range equalizer graphic — a bar from 5 to 30 reps all marked "builds muscle IF near failure," with the practical 6–15 zone highlighted and labeled "for joint and time reasons"

Strength ≠ size

You can gain strength without gaining much muscle (skill and nervous-system adaptation), and you can build muscle across a wide load spectrum. The logbook moving up is still your best long-term progress signal — but the *mechanism* of growth is tension applied to the fiber, not the number on the bar.

What this means in one sentence

> **Pick exercises you can load hard and control well, take most sets within 1–3 reps of failure at roughly the rep counts that suit the exercise, and progress over time. Everything else in this guide is detail on top of that.**

Chapter 2 — Where the Experts Disagree (and what it means for you)

Most fitness guides give you one guru's opinion and pretend the debate is settled. I'm not going to do that, because the debate *isn't* settled — and knowing where the experts genuinely disagree makes you a smarter lifter than 95% of gym-goers.

I built this guide from 40+ coaches, researchers, and educators. Here are the real fault lines.

Fault line 1: How close to failure?

- One camp: most sets at 2–3 RIR is plenty; failure adds fatigue faster than growth.
- Another: growth measurably spikes when sets hit 0 RIR vs 2 RIR (a 2022 meta-analysis), though a second 2022 meta-analysis found no advantage.
- The outlier: one hardline natural coach rejects RIR entirely — "train to failure, period" — and credits his best growth to it.

The synthesis that survives: *the exercise decides.* Heavy free-weight compounds (squat, bench, deadlift) cost disproportionate recovery when taken to failure — keep 1–3 RIR even on the last set. Machines, cables, and isolations are cheap to fail on — take the last set all the way.

Fault line 2: How much volume?

- Mainstream: 10–20 hard sets per muscle per week for growth; up to 20–30 for smaller muscles.
- The weekly-stimulus framework: past 10–12 sets/week, returns become minuscule relative to fatigue; 6–8 quality sets can out-grow sloppy 20s.

Context explains most of the gap: the high-end numbers come from elite and chemically-assisted bodybuilding, where recovery capacity is a different sport. If you're a natural intermediate, treat 10–20 as the ceiling of a range you *earn*, not a target you *owe*.

Fault line 3: Bro splits — useless or fine?

- Critics: once-per-week frequency plateaus advanced trainees; distributing volume across 2–3 sessions gives higher-quality sets.
- Defenders: if effort and total volume are there — and you *love* the split — it works, especially for people who train with high per-session intensity.

The synthesis: frequency is mostly a distribution tool. 16 sets in one session decay in quality; 8+8 across two days doesn't. And the split you'll actually show up for beats the one you won't — every source agrees on that.

IMAGE

"The Fault Lines" summary card — 3 rows (failure / volume / bro split), each with the two camps' one-line positions and the synthesis verdict

The meta-lesson

When experts disagree, the disagreement almost always traces to context: natural vs assisted, novice vs advanced, high recovery budget vs low. Whenever you read a training claim, ask *"for whom?"* — that one question filters most of the confusion in this industry.

Chapter 3 — Progression: The Only Non-Negotiable

Every effective program in history is the same program underneath: a vehicle for **progressive overload** — doing slightly more over time. It's the progression from week to week that drives results, not monumental effort in any single workout.

Overload is more than adding weight

When people hear "progressive overload" they think "add 5lbs to the bar." That's one route. The full list:

- More **reps** at the same weight
- More **sets**
- **Better technique** — tighter control, fuller range, deeper stretch
- Better **mind-muscle connection** (real for isolations especially)
- A **better exercise swap**

All five count. All five move the needle. Only one of them shows up on the bar.



Double progression: the workhorse method

The classic, used for a century because it works:

1. Pick a rep range — say 8–12 — and a weight you can do for 8.
2. Add reps session by session until you hit 12.
3. Add weight, drop back to 8, repeat.

Simple, self-regulating, and it works on nearly everything.

Block structure: how weeks fit together

Random double progression works for a long time, but structured programs organize it into **blocks** — typically 3–6 weeks of overload followed by a deload week. Within a block you can progress by:

- **Adding reps weekly** (log every set; expect a natural drop-off like 9/8/7/6 across sets)
- **Adding sets weekly** (chest 14 → 15 → 16 → 17 across the block)
- **Ramping effort weekly** (RPE 6 → 7 → 8 → 9; reps rise as you close in on failure)

Between blocks, *something* changes — even if it's tiny: 1–2 more weekly reps, +5% load with a lower rep range, an exercise swap every 2–5 blocks when a movement stalls below ~6 reps. "If you do the optimal thing year-round, it's no longer optimal".



The logbook: your only judge

Here's the feedback loop that separates lifters who progress from lifters who guess:

> **Next session's numbers didn't move? The previous session's volume was too high — cut it and rerun.**

> **Numbers went up? You found your sweet spot. Keep it, and nudge upward.**

Not soreness. Not the pump. Not how "hard it felt." **The logbook moving up over months is the only progress signal that matters**.

Chapter 4 — Sets, Reps, and Rest

How many sets?

The dose-response curve for volume is an inverted U: the first sets of the week do the most, returns diminish, and eventually fatigue wins.

Working landmarks, per muscle, per week:

Weekly sets	Effect
~3	Maintenance / retention
6–8	Steady growth on minimal volume

10–20	The standard growth prescription
20–30	High-responder / assisted territory — trial, don't assume

And within a single session, stimulus plateaus around **~6 hard sets per muscle** — which is why 16 sets split across two days beats 16 sets crammed into one. Per exercise, cap at about 5 sets before switching to a different angle.

IMAGE

Inverted-U curve — sets/week on the x-axis, growth on the y-axis, maintenance/standard/diminishing zones marked

Which rep counts, practically?

Since 5–30 reps grow muscle equally near failure, choose by *suitability*:

- **Compounds: 6–10** — heavy enough to be efficient, not so heavy that joints and setup dominate.
- **Isolations: 10–15 (or higher)** — light enough for joint comfort, hard enough to hit failure cleanly. Curls under 10 reps tend to irritate elbows; squats above 12 become breathing-limited before leg-limited.

Rest: longer than you think

2–3 minutes between sets is the hypertrophy baseline — and 3 minutes beats 1 minute. Heavy leg work may need 4–5. That feels like a lot if you're used to 60-second bro-rest, but performance per set — and therefore stimulus per set — is what you're resting for.

Time-saving option that *doesn't* cost performance: **antagonist supersets** — biceps with triceps, quads with hams — one muscle recovers while the other works. And your rest capacity is trainable: gradually shortening rest over weeks costs surprisingly little growth.



Frequency: less critical than you've been told

With weekly volume equalized, exact frequency barely moves the needle — it's a logistics and enjoyment decision. Practical consensus: hit each muscle **at least twice a week**, mostly because it distributes volume into higher-quality sets. One deep-dive argues low-volume/high-frequency is the true sweet spot for advanced naturals — less damage per session, more of each week's recovery spent building. You don't need to pick a side: 2×/week sits comfortably in the overlap.

Chapter 5 — Exercise Selection

No mandatory exercises exist

The muscle doesn't know the exercise's name — it knows the tension you deliver to it. There is no lift you *must* do, only lifts that fit your mechanics, goals, and equipment.

The first filter is always **pain**: any exercise that hurts a joint gets modified or swapped, immediately, no exceptions.

The second filter is **stimulus-to-fatigue ratio** — the most stimulus for the least fatigue. A deadlift is a massive stimulus but a massive fatigue bill; a chest-supported row or machine press delivers comparable targeted tension at a fraction of the systemic cost.

Machines deserve more respect

Research keeps finding machines equal to — possibly better than — free weights for hypertrophy: smooth resistance curves, tension in the stretched position, and safety when a set goes to failure. Free weights still earn their place: accessibility, carryover, stabilizer work. But "free weights are for real lifters" is ego, not evidence — and the failure-training guidance from Chapter 2 works precisely *because* machines make failure cheap.



Choose exercises that load the stretch

The lengthened position of a rep is the most hypertrophic part of it — the deep stretch beats the squeeze:

- Preacher or incline curls (tension at the bottom) over standing curls
- Full squats over shallow squats for quads
- Deep-stretch presses and pull-ups
- Leg press feet low and narrow for quads; high and wide shifts to glutes/hips

If an exercise has no meaningful stretch, fix the depth, fix the mobility, or swap the exercise.

Ordering and volume across exercises

- **Compounds first, isolations later** — with one exception: whatever you're *prioritizing* goes first while you're fresh.
- About **5 sets max per exercise per session**; more volume for a muscle → add another angle. Larger muscles genuinely benefit from two angles (e.g., vertical + horizontal pulling).
- Warm-up sets don't count as volume — 1–2 light ramp sets before the first exercise per muscle, then the real work begins.

Overlap is free volume

Muscles share work: pressing trains the medial/lateral triceps heads and front delts; pulldowns with full ROM even feed the pecs. This is why biceps, side delts, and rear delts — the muscles pressing *doesn't* cover — are the ones that genuinely need direct isolation.

Chapter 6 — Choosing Your Split

Everything from Chapters 1–5 is the **engine**. The split is just the **container** you run it in — and that reframe alone kills half the internet's training arguments.

The decision tree

Work top to bottom; the first YES wins:

```
Training age < ~1 year?
```

```
  → Full Body 3×/week (a little stimulus does a lot; no deloads needed yet)

Sessions running past ~1.5h, or back-half lifts stalling from fatigue?

  → Move up the ladder: Full Body → Upper/Lower → PPL

Fresh at the end of every session?

  → You can move DOWN the ladder and concentrate stimulus quality —

    regression can be progress

Only ~45 minutes per session?

  → Body-part split days (max 2 muscle groups per day)

Schedule can't commit to fixed weekdays?

  → Rotating PPL (3 on, 1 off), or hybrid PPL with fixed rest days

Arms (or one muscle) badly lagging?

  → Specialization: cut a strong area to ~3 maintenance sets,

    reallocate that volume to the priority at 2–3×/week

Healthy intermediate, no complaints?

  → Upper/Lower, 4 days — the intermediate default
```

IMAGE

The decision tree as a designed flowchart card — the signature graphic of the whole guide; a condensed version becomes the free cheat sheet

The meta-rules every source agrees on

1. **Volume needs dictate the split — never pick the split first and force volume into it**.
2. **The best split is the one you'll actually run.** Consistency beats theoretical optimality, every time.
3. **Audit before you switch.** A new split that quietly *reduces* your per-muscle volume is an atrophy program.
4. **If you're progressing on everything, change nothing**.

Chapter 7 — Recovery, Fatigue, and Deloads

Training is the stimulus. Recovery is where you actually grow. Under-recover and it doesn't matter how good the program is.

Scheduling basics

- Leave **48–72 hours between hard sessions for the same muscle**.
- Put your hardest sessions **after rest days**.
- Deadlifts are the biggest systemic drain — schedule them before your most rest days.
- Never more than 3 consecutive training days without a rest day in a body-part split.

The warning signs of under-recovery

Learn these and you'll never need to guess:

- Weights that used to feel light feel heavy
- Soreness that lingers longer than usual
- No pump, even on isolation work
- Losing motivation to train; feeling run down; getting sick often

Any cluster of these means intervention time — not "push through."

Deloads: match the medicine to the illness

A deload is an easy week that dissipates accumulated fatigue. Two rules make them work:

1. **Match the lever to your program's stressor:** volume-driven program → cut sets ~50%, keep loads; intensity-driven → cut loads to ~60%, keep sets. Nothing above 90% ever, on any deload.
2. **It should feel easy.** If your light week still feels hard, you timed it perfectly — that's fatigue leaving.

The foolproof variant I use with clients: the **frequency deload** — change nothing about the workouts, just spread the week across 8–12 days.

How often? Roughly every 8–12 weeks, or by symptoms — and honest novices usually don't need them at all, because they haven't accumulated enough stress yet. One final guardrail: **~10 days fully off is where detraining begins** — full breaks are fine a couple of times a year, but don't turn a deload into a sabbatical.

IMAGE

Deload decision card — "cut volume 50% / keep loads" vs "cut loads to 60% / keep sets" vs "frequency deload: stretch the week," plus the symptoms checklist

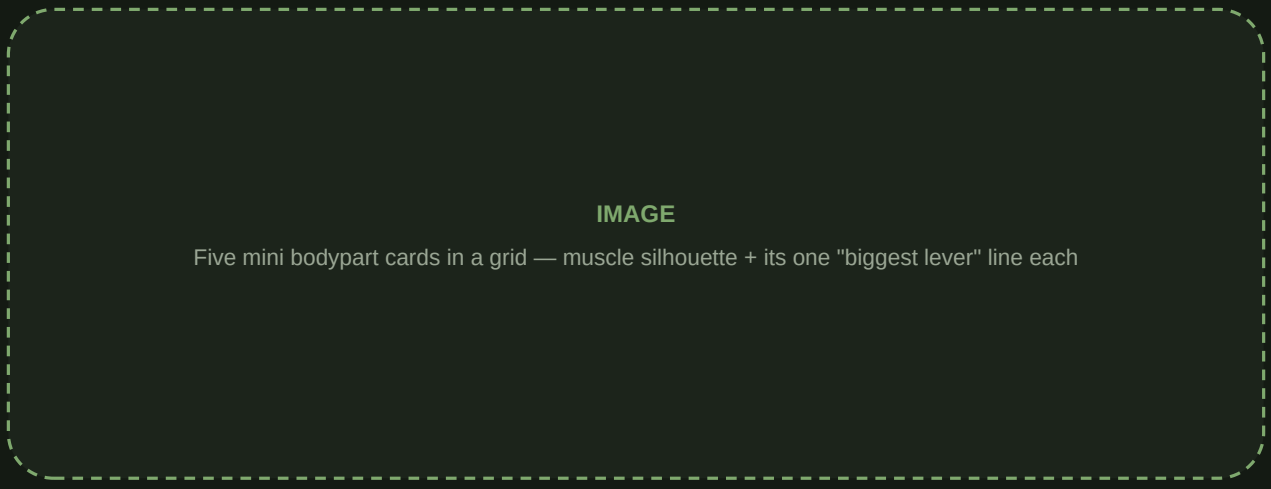
The unsexy 80%

Adequate protein (Part 2), sleep, and not letting life stress stack on top of training stress — every source lists these as non-negotiable. No recovery trick replaces them.

Chapter 8 — Bodypart Notes: The Shortcuts

One page each — the highest-value, most-agreed-on guidance per muscle, pulled from the full muscle-specific research in my knowledge base. (These condense dozens of hours of coaching material.)

- **Back:** Rows at least match pulldowns for growth; mix vertical + horizontal pulling; watch the >25 weekly sets recovery cliff; grip choice shifts width vs thickness emphasis.
- **Chest:** Press from multiple incline angles; machines/Smith presses earn their place; 2×/week beats one marathon session; individual sternum angle explains why your buddy's favorite incline does nothing for you.
- **Legs:** Train hamstrings (curl-first ordering works), quads through full depth, and don't expect gym calves to out-grow what your daily life load does — dedicated stretch work is the lever.
- **Shoulders:** Side delts are the visual priority for most (12–16 sets/week over 3 days); rear delts mostly come along with good rowing — direct work is the garnish; pressing covers front delts amply.
- **Arms:** 3–6 direct sets/week is a real starting point because back and pressing work covers plenty; long-head triceps (overhead extensions) and biceps stretch work (incline/preacher curls) are the growth levers; specialization phases raise arms to 9–14 sets only when everything else is settled.



End of Part 1. → Part 2: Nutrition.

Part 2 — Nutrition

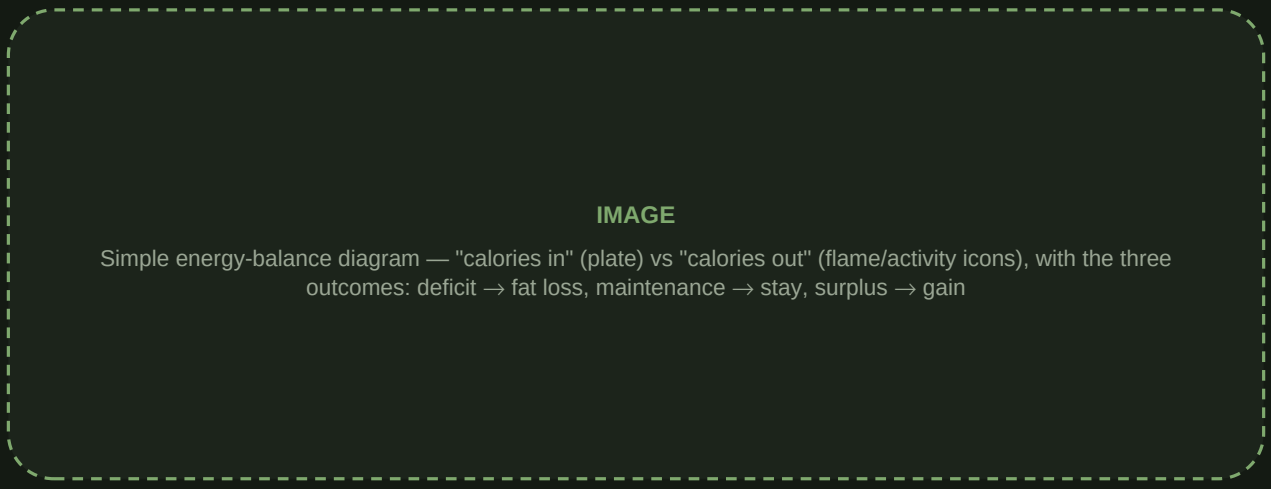
Chapter 9 — Calories: The Lever That Decides Everything

You can't out-train a bad diet, and you can't out-supplement a bad diet either.

Every physique goal you could possibly have — losing fat, building muscle, recomposition — starts with one number: your total daily calorie intake. Not how "clean" you eat. Not your meal timing. Not skipping carbs after 6pm. Total calories decide whether your body is in a position to lose weight or gain it. Everything else in this part is fine-tuning on top of that.

What a calorie actually is

A calorie is simply a unit of energy — the energy contained in the food you eat and stored in the tissues of your body. Your body burns through that energy all day long: keeping your organs running, fueling your brain, digesting food, and powering your training.



Fat loss requires a deficit

To lose fat, you need to consume fewer calories than you burn. When you do, your body is forced to pull from its stored energy — your body fat — to cover the shortfall. That's the entire mechanism. There is no way around it and no trick that bypasses it.

Muscle growth (usually) requires a surplus

To build muscle as effectively as possible, you need to consume more calories than you burn. Building new tissue is energy-expensive work, and training hard on fumes doesn't support it.

There are two exceptions where you can build muscle in a deficit:

- **If you're carrying higher body fat** — the energy stored in fat tissue can help fuel muscle growth.
- **If you're a novice or returning from a layoff** — your body builds muscle readily in those windows regardless of calorie balance.

If neither applies to you and maximum muscle is the goal, you'll need a small surplus. How small? That's coming up.

Step 1: Find your maintenance calories

Your maintenance level is the intake at which your weight stays roughly stable. There are two practical ways to estimate it.

Method 1 — Track what you already eat. If your weight has been stable and your diet is fairly consistent, track a normal week of eating (an app makes this painless) and average it out. That average is your approximate maintenance. This is the more accurate method because it's built from your real life, not a formula.

Method 2 — Use a calculator. If you know your body fat percentage, the Katch-McArdle formula is the better pick. If you don't, use Harris-Benedict. No calculator is perfect, but either gets you close enough to start — you'll correct from real-world data anyway (next section).

IMAGE

Two-panel graphic — "Method 1: track a week" (phone with food log) vs "Method 2: calculator" (simple formula card), both pointing to "your starting number"

Step 2: Apply your goal

If your goal is fat loss:

- A **500-calorie deficit** below maintenance is the sweet spot for most people — roughly one pound of fat lost per week (a pound of fat holds about 3,500 calories).
- If you can handle the extra hunger and want to move faster, **750 below** works.
- Experienced dieters can run **1,000 below** for short bursts, but the leaner you get, the more aggressive deficits cost you training performance and adherence.

If your goal is muscle gain:

- **100–200 calories above maintenance.** That's it. Beginners can sit toward the higher end; advanced lifters who want to stay lean should sit toward the lower end.
- There is no prize for bulking harder. Muscle growth is capped by how much tissue your body can actually synthesize — past that cap, every extra calorie is stored as fat. All an aggressive bulk buys you is a longer, hungrier cut later.

IMAGE

Number-line graphic — deficit tiers on the left (–500 / –750 / –1000 with "who each is for") and surplus window on the right (+100 to +200)

Step 3: Track your progress and adjust

Your starting number is a hypothesis. Real-world data tells you what to do with it.

Weigh yourself daily, first thing in the morning, and work with the **weekly average** — daily weight swings are water, food residue, and noise; the weekly trend is the truth.

- **Cutting:** aim to lose about 0.5–1% of your body weight per week. Losing nothing for 2–3 weeks? Drop another 100–150 calories (or add a bit of daily activity) and reassess.
- **Lean gaining:** aim for about 0.5–1.5 lb per month of total weight gain (beginners at the top end, advanced at the bottom; women roughly half these figures). Gaining much faster than that means the surplus is mostly going to fat. Gaining nothing for a month? Add 100–150 calories.

This adjust-and-reassess loop is exactly what I do with coaching clients every week — it's simple, but it's the difference between a plan that works on paper and one that works on you.

IMAGE

Simple decision flow — "weigh daily → weekly average → on-pace? yes: keep going / no: adjust ±100–150 kcal, reassess in 2 weeks"

One warning before you chase "shredded"

Staying super lean year-round sounds appealing, but for most people it means under-fueled training and stalled muscle growth. Some fat gain during a building phase is normal and expected — modest is the goal, not zero.

Chapter 10 — Protein First

Calories decide whether your weight changes. Protein decides *what kind* of weight changes.

Protein provides the amino acids your body uses to build and repair muscle tissue. Skimp on it while training hard, and you're doing the work without supplying the raw materials.

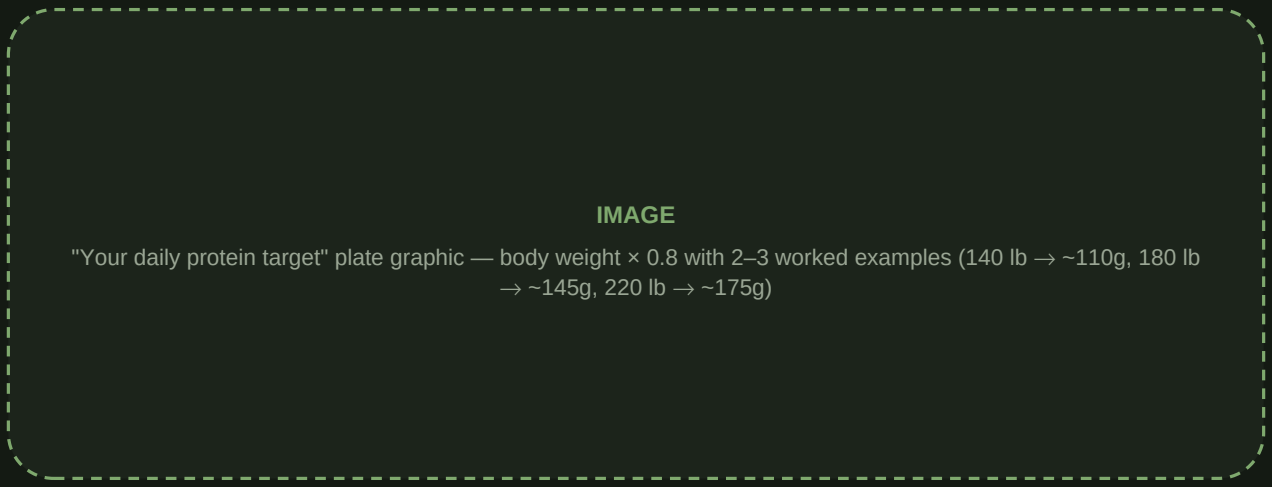
How much do you actually need?

Less than the supplement industry would like you to believe — but probably more than you're currently eating.

Research shows around **0.7g per pound of body weight per day** is enough to maximize muscle growth for most people. To sit comfortably on the safe side:

> **Aim for 0.8g of protein per pound of body weight daily (about 1.8g/kg).**

Going higher is fine if you enjoy protein-rich foods, but past this point it's unlikely to add anything except fullness — which, to be fair, is genuinely useful during a cut.



Quality and sources

Animal-based proteins are the most effective gram for gram — they're complete (all essential amino acids) and highly bioavailable. The usual suspects:

- Poultry
- Beef
- Eggs
- Fish and seafood
- Pork
- Dairy (Greek yogurt and cottage cheese are workhorses)
- Protein powder (a food, not a magic — see Part 3)

If you're plant-based: you can absolutely hit your target, it just takes slightly more planning. Prioritize legumes, tofu, tempeh, seitan, and a plant protein powder; eating a variety across the day covers the amino acid spectrum. Aim for the same total — don't inflate it with bro-science math.

And for everyone: the small amounts of protein in oats, bread, rice, and vegetables **count toward your total**. You don't need every gram from "perfect" sources.

Splitting it across the day

Total daily protein is the main thing, but if you want to leave nothing on the table, spread it across **at least three feedings** of roughly 0.3–0.4g per kg each. Two feedings works nearly as well; one giant daily dose still does most of the job. This is fine-tuning, not pass/fail.



Hitting protein on a busy schedule

This is where clients struggle most, so here's the coaching shortcut list:

1. **Anchor every meal with a protein source first**, then build the rest of the meal around it.
2. **Default to double portions** of the protein you'd normally serve yourself.
3. **Keep two zero-effort options stocked** — e.g., Greek yogurt tubs and a protein powder you like.
4. **When eating out, choose the meal by its protein.** Most menus have a chicken, fish, or steak option.

None of this is exotic. It just front-loads the one macro that matters most.

Chapter 11 — Carbs and Fat: The Support Crew

Once protein and total calories are set, fat and carbohydrates fill in the remaining energy. There's no magic ratio between them — there are just sensible ranges and personal preference.

Fat: don't fear it, don't drown in it

Dietary fat got unfairly demonized for years. It's essential for hormone production (including testosterone), brain function, and absorbing certain vitamins.

> **A practical range: 15–30% of total calories from fat.**

That's enough to cover its jobs while leaving room for protein and carbs. You can run slightly below 15% for a short, aggressive cut, or sit higher if you genuinely prefer fattier meals — most people

land in range without specifically tracking it.

Favor these sources:

- Nuts and nut butters
- Seeds
- Avocado
- Olive oil
- Fatty fish

Saturated fat takes care of itself in a varied diet; you don't need to hunt for it.

Carbs: the training fuel

Carbs are not essential for survival — but they're essential for training hard and building muscle at your best. They're stored in your muscles as glycogen, which is your primary fuel for intense sets. Cut carbs hard and, for most people, both performance and mood go with them.

Carbs have also been wrongly blamed for fat gain via insulin. The honest version: insulin is a storage hormone, its rise after carbs is normal and temporary, and net fat gain still comes down to total calories. Carbs don't make you fat — a calorie surplus does.

> **Practical approach: set protein and fat first, then let carbs fill the remaining calories.**

Where your carbs should come from

All carbs end up as glucose (plus a little fructose), so in that sense a carb is a carb. The difference is the packaging — what comes *with* the carbs:

- **Build the base on minimally processed whole foods:** fruit, vegetables, potatoes and sweet potatoes, oats, rice, beans and legumes, whole grains. These bring fiber, micronutrients, and fullness.
- **Treat processed carbs as the garnish, not the meal.** If roughly 80–90% of your intake is nutrient-dense whole food, the other 10–20% can be whatever you actually enjoy — within your calories, it costs you nothing in progress.



Fiber — the unglamorous number that matters

Buried at the end of most diet advice, but your digestion and long-term health run on it: aim for roughly **14g of fiber per 1,000 calories eaten** (ballpark 25–35g a day for most). If you're building your diet from the whole-food carb list above, this mostly handles itself.

Carb timing (the short version)

Total daily carbs matter far more than timing — but if you want a small edge, place a decent portion of them in the meals **before and after training**. Fuel for the session, recovery after it. That's the whole strategy.

Chapter 12 — Meal Timing & Frequency: Mostly Noise

This is the most over-hyped corner of fitness nutrition. You've heard the rules:

- "Eat every 2–3 hours to stoke your metabolism."
- "Get a shake within 30 minutes of training or the workout is wasted."
- "No carbs after 6pm."

Here's the truth: **total daily calories and protein dominate outcomes.** Meal frequency and timing are fine-tuning worth a few percent at most.

How many meals?

Whatever lets you hit your numbers consistently. Some people thrive on two big meals; others need five smaller ones to manage hunger. Both work. The "stoking the metabolism" effect of frequent eating is negligible — and for fat loss specifically, meal frequency makes no measurable difference at equal calories. Eat in the pattern that makes your diet *easy to sustain*, because sustainable is what actually produces results.

The one guideline worth keeping from Chapter 10: if maximum muscle is the goal, get at least three protein feedings across the day.

Before training

No perfectly engineered pre-workout meal exists. A basic protein + carb meal a few hours before training covers it: fueled, not weighed down. Examples: chicken and rice, eggs and toast, yogurt

and fruit with granola.



After training

The "anabolic window" is far more forgiving than advertised. Protein digestion is slow — the amino acids from your pre-workout meal are still circulating when you rack your last set. Just get a quality protein feeding within a couple of hours after training, with carbs if you like (a typical lifting session only drains a modest amount of glycogen, which your normal diet restocks easily). Shake or whole food — whichever fits your routine.

The overnight "fast"

Nothing bad happens overnight. Your body doesn't flip into a muscle-wasting state because you slept 8 hours without protein. Handle your total daily protein, and the rest is scheduling.

Chapter 13 — Eating Around Your Real Life

Most diet advice assumes you live in a vacuum: same schedule, same kitchen, no travel, no social life. You don't, and your diet has to survive contact with reality or it won't last. This is the chapter most guides skip.

The 80/20 rule is a lifestyle rule

Carry the logic from Chapter 11 beyond carbs: if the large majority of your eating is on-plan, the minority can flex. One restaurant dinner doesn't undo a week; one perfect week doesn't earn a lost weekend. Consistency over months is the only currency that spends.

Travel and hotel gyms

- Default to the protein-first rule when ordering out (Chapter 10).
- Keep breakfast boring and repeatable — it's the easiest meal to automate (e.g., yogurt, fruit, coffee).
- Hotel days are maintenance days at worst. A week of approximate eating costs you almost nothing if the surrounding weeks are consistent.

Busy weeks and shift work

If your schedule is chaotic, shrink the decision surface:

- Two or three default meals you can shop for and cook on autopilot.
- A short list of acceptable grab-and-go options near where you actually spend your days.
- Protein powder as the emergency backstop — not because it's magic, because it's a 30-second meal insurance policy.

Diet breaks and being human

Leaner, longer cuts benefit from planned breaks at maintenance — a week or two eating at maintenance to reset hunger and psychology, then back to the deficit. Not mandatory, but a useful tool. And if life happens and you fall off entirely: the plan isn't ruined. You don't "restart Monday." You just eat the next meal on-plan.

IMAGE

"Real life" cheat-sheet card — travel / busy week / social meal, one practical rule each

End of Part 2. → Part 3: Supplementation.

Part 3 — Supplementation

Chapter 14 — Supplements: The Honest Hierarchy

Let's set expectations before anything else:

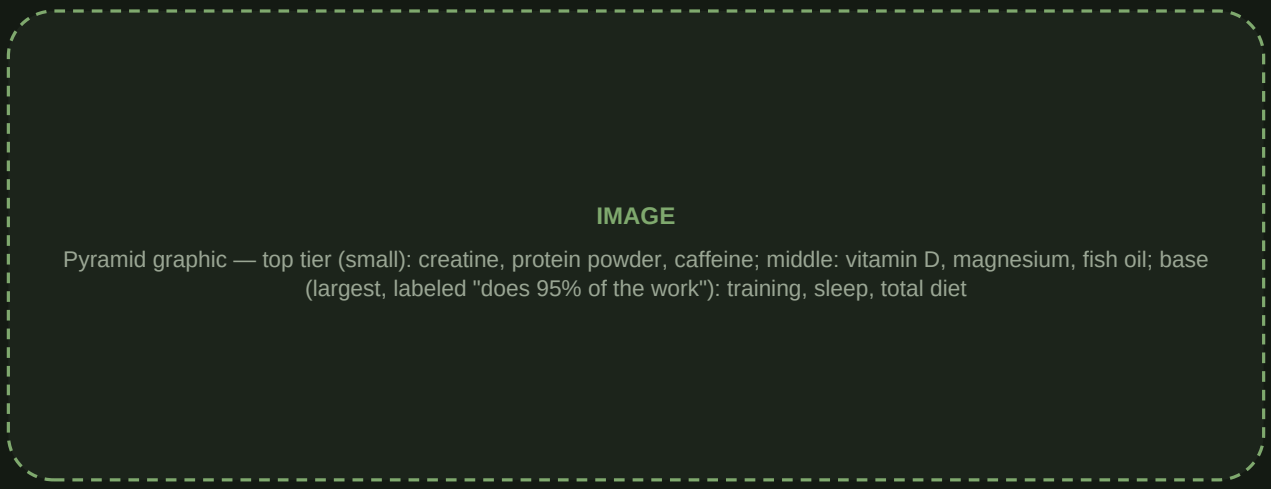
> **Supplements don't fix anything. Consistent training and a sensible diet fix things. Supplements are small multipliers on top of work that's already being done.**

If your training is random and your diet is guesswork, no pill or powder will save you — and anyone selling you one is selling hope, not results.

But once training and nutrition are in place, a short list of supplements earns its spot: they make hitting protein targets easier, fill genuine nutritional gaps, and give you a measurable (if modest) edge in performance and recovery.

The market is mostly noise

Walk into any supplement store and you're looking at a wall of products built on underdosed formulas, unproven ingredients, and marketing budgets larger than their research budgets. The good news: the list of supplements with real evidence behind them is **short**, which makes this part of the guide short too. That's a feature.



How to read a label like a coach

Before the individual breakdowns, four filters that catch most of the junk:

1. **Dosed transparently?** Proprietary blends hide amounts behind a fancy name. If you can't see the dose of each ingredient, you can't trust the product.
2. **Dosed high enough?** The label lists an ingredient that looks great at 12mg and does something at 3,000mg. Underdosing is the industry's favorite trick.
3. **Few ingredients, or twenty?** Long formulas usually mean many things dosed at nothing. More words is not more effect.

4. **Claims vs reality.** Anything promising dramatic transformation is telling you exactly how little it does.

Chapter 15 — What Actually Works

Creatine monohydrate — the king

The most researched sports supplement in existence, with thousands of studies behind it: better training performance, more lean mass over time.

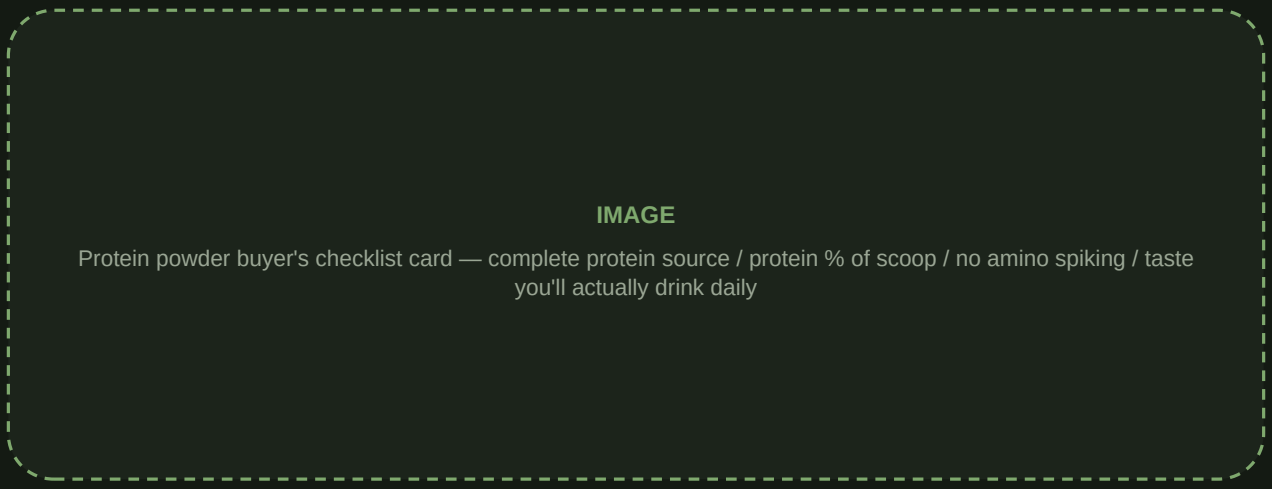
- **Monohydrate only.** Every "advanced" form (HCl, ethyl ester, buffered) matches it at best and costs more at worst.
- **3–8g daily**, scaled roughly to body weight. Any time of day, any drink you like.
- **No loading needed.** Loading saturates your stores a week or two faster; the destination is identical.
- **No cycling needed.** Your body doesn't build tolerance; there's nothing to cycle off from.
- It's cheap. If you buy one supplement, buy this one.



Protein powder — food in convenient form

Not a magic muscle-builder — a convenient way to hit the protein targets from Chapter 10.

- **Timing doesn't matter.** Pre, post, breakfast, bedtime — whatever fits your day.
- **1–3 scoops a day** is the realistic working range for most people.
- **Check for amino spiking:** if the ingredients list includes individual amino acids (glycine, taurine, alanine) alongside the protein, the company is inflating the protein number on the label cheaply. Skip it.
- **Choosing one, simply:** pick whey (concentrate is fine; isolate if dairy sits heavy or you want fewer carbs/fat per scoop), check the protein-per-serving-to-weight ratio, and make sure you like the taste enough to drink it daily. A plant blend (pea + rice) works fine if dairy isn't for you.



Caffeine — the real pre-workout

Most of what a commercial pre-workout does for you is the caffeine. The rest is usually underdosed decoration.

- **3–6mg per kg of body weight**, 20–30 minutes before training, for a genuine boost in strength, focus, and endurance.
- **Simplest version that works:** a strong coffee. Same effect, fraction of the price.
- **Protect your sleep:** nothing caffeinated within 6–8 hours of bedtime — sleep does more for your progress than any pre-workout ever will.
- If you train in the evening, caffeine-free pre-workout is mostly unnecessary; a proper day of eating is your pre-workout.

Vitamin D — the one worth considering

Many people run low, especially through winter or indoor-heavy lives, and it matters for training, mood, and general health.

- A daily dose in the 1,000–2,000 IU range is the common general starting point; a blood test turns guesswork into a real number.
- Sunshine is the original source — supplementing covers the gap.

Magnesium — for the hard-training and the stressed

Involved in hundreds of processes, and training increases your needs. Many people fall short.

- Look for **magnesium glycinate or citrate** (well-absorbed); oxide is cheap and poorly absorbed.
- Evening dose pairs nicely with sleep, where it may modestly help.

Fish oil — conditional, not compulsory

If you eat fatty fish (salmon, sardines, mackerel) 2–3 times a week, you likely don't need it. If you don't, an EPA/DHA-based product covers that gap. Check the **combined EPA+DHA per serving**, not the total oil weight — that's where labels play games.

Multivitamin — optional insurance

A multi can't fix a bad diet, but as a low-cost backstop for micronutrients that are hard to get from food alone (vitamin D, magnesium, zinc, K are the usual suspects for lifters):

- Prefer formulas with **few ingredients at meaningful doses** over kitchen-sink labels with 30 items at homeopathic doses.
- Think of it as the spare tire, not the engine.

IMAGE

Final "stack" summary card — creatine daily + whey as needed + caffeine pre-training, with the conditional row (vitamin D / magnesium / fish oil / multi) beneath

The Closing Word on Supplements

Notice what's *not* in this part: fat burners, testosterone boosters, BCAAs, detox teas, and every powder promising a shortcut. That silence is the advice.

Your results will come from the training principles in Part 1 and the nutrition framework in Part 2 — executed consistently for months. A tub of creatine and a bag of whey quietly support that work. Nothing replaces it.

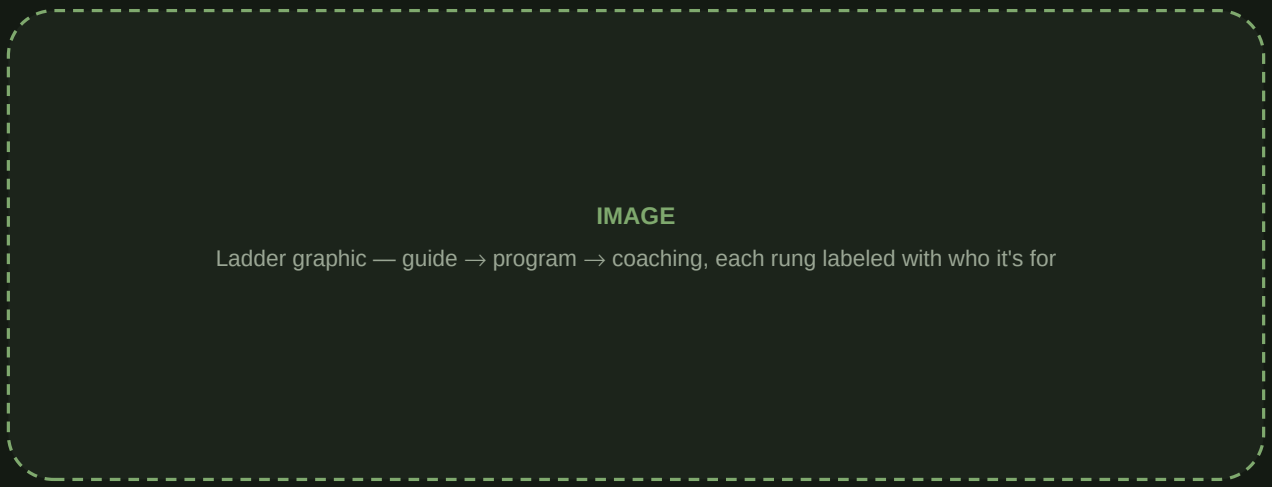
End of Part 3. → Outro: Your Next Step.

Your Next Step

You now know more about building muscle than the person doing their fourth exercise-swap of the month ever will. Knowledge was never the hard part, though — **execution over months is**, and that's where structure wins.

Here's the ladder, depending on where you are:

1. **Pick your split.** Re-read the decision tree in Chapter 6 and match it to your schedule — not your ambition.
2. **Run the engine.** Log every working set. Progress reps, then weight. Let the logbook judge the program, not your feelings.
3. **When you want it done for you** — a fully built program to run yourself — my Training Programs are built on this exact system. (Coming soon.)
4. **When you want it built around you** — custom training and nutrition, weekly check-ins, unlimited messaging, all centralized in my coaching app — that's 1:1 coaching, and you can apply at pingliftsfitness.com.



Whichever rung you start from: train close to failure on things that load the stretch, eat enough protein and the right calories, sleep, and let the logbook climb. Do that for a year and you won't recognize yourself.

Discipline and consistency beat shortcuts. The results that matter are the ones that last.

— Ping

IMAGE

Closing photo — you posing/competing (WNBF Pro card context), dark/green treatment, site URL

General fitness information for educational purposes — not medical advice. Consult a professional before starting any training or nutrition program.

